MATH1025B
Essential Mathematical Methods

# Chapter 5: Sequences

October 2, 2025

### Definition 5.1 (Sequences)

A sequence of real numbers is a function $a : \mathbb{Z}^+ \to \mathbb{R}$.
Given $n$, the function returns the $n$-th term, which is a real number.
Usually, we write $a_n$ instead of $a(n)$ and we denote the sequence by $\{a_n\}$.

### Example 5.1

(a) $a_n = (-1)^n$, the sequence is $-1, 1, -1, 1, \ldots, (-1)^n, \ldots$

(b) $a_n = 2^{n-1}$, the sequence is $1, 2, 4, 8, \ldots, 2^{n-1}, \ldots$

## Example 5.1

(c) Let $f(x) = \sin x$. Then

$$f^{(n)}(x) = \begin{cases} \cos x & \text{if } n = 4m+1 \\ -\sin x & \text{if } n = 4m+2 \\ -\cos x & \text{if } n = 4m+3 \\ \sin x & \text{if } n = 4m \end{cases} \qquad f^{(n)}(0) = \begin{cases} 1 & \text{if } n = 4m+1 \\ 0 & \text{if } n = 4m+2 \\ -1 & \text{if } n = 4m+3 \\ 0 & \text{if } n = 4m, \end{cases}$$

where $m$ is a nonnegative integer.
Let $a_n = f^{(n)}(0)$. How do we write down the general term?

$$a_n = \cos\left(\frac{(n-1)\pi}{2}\right)$$

or

$$\begin{aligned} a_n &= \text{ real part of } i^{n-1} \\ &= \frac{1}{2}\left[i^{n-1} + (-i)^{n-1}\right] = \frac{i^{n-1}}{2}\left[1 + (-1)^{n-1}\right] \end{aligned}$$

## Exercise 5.1

Write down the general term of each of the following sequences

(a) $1, 0, 1, 0, \ldots$

(b) $2, 2 \cdot 4, 2 \cdot 4 \cdot 6, 2 \cdot 4 \cdot 6 \cdot 8, \ldots$

(c) $1, 1 \cdot 3, 1 \cdot 3 \cdot 5, 1 \cdot 3 \cdot 5 \cdot 7, \ldots$

(a) **Ans:** $\frac{1}{2}[1 + (-1)^{n-1}]$

(b) **Ans:** $2 \cdot 4 \cdots 2n = 2^n n!$

(c) **Ans:** $1 \cdot 3 \cdots (2n-1) = \frac{1 \cdot 2 \cdot 3 \cdots 2n}{2 \cdot 4 \cdot 6 \cdots 2n} = \frac{(2n)!}{2^n n!}$

A **recursive sequence** is a sequence where each term is defined using one or more preceding terms.

## Example 5.2

Consider the sequence $\{a_n\}$ defined by $a_1 = 4$ and $a_{n+1} = 2a_n + 1$ for $n \geq 1$. The recursive relation tells us how to find $a_2, a_3$, and so on.

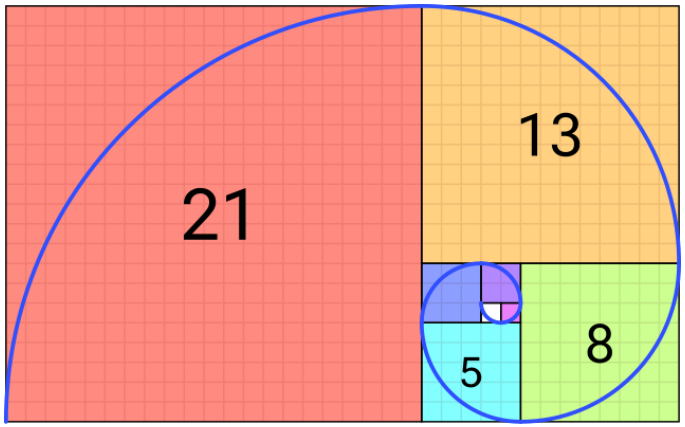
$$\begin{aligned} a_2 &= 2a_1 + 1 = 2(4) + 1 = 9, \\ a_3 &= 2a_2 + 1 = 2(9) + 1 = 19, \\ a_4 &= 2a_3 + 1 = 2(19) + 1 = 39, \ldots \end{aligned}$$

It turns out the general term of $\{a_n\}$ is

$$a_n = 5 \cdot 2^{n-1} - 1 \quad \text{for } n \in \mathbb{Z}^+.$$

## Example 5.3 (Fibonacci Sequence)

The sequence $\{F_n\}$ defined by $F_1 = F_2 = 1$ and $F_{n+2} = F_{n+1} + F_n$ for $n \geq 1$ is called the **Fibonacci sequence.**




The Fibonacci Spiral
https://en.wikipedia.org/wiki/Fibonacci_sequence

## Exercise 5.2

Let $\varphi = \dfrac{1+\sqrt{5}}{2}$ ($\approx 1.618$ called **Golden Ratio**) and $\psi = \dfrac{1-\sqrt{5}}{2}$.

Show that $F_n = \dfrac{\varphi^n - \psi^n}{\varphi - \psi}$ for all positive integer $n$.

### Example 5.4

Let $\{a_n\}$ be a sequence defined by $a_1 = 1$ and $a_{n+1} = a_n^2 + 1$ for $n \geq 1$. Then,

$$a_2 = a_1^2 + 1 = 1^2 + 1 = 2,$$
$$a_3 = a_2^2 + 1 = 2^2 + 1 = 5,$$
$$a_4 = a_3^2 + 1 = 5^2 + 1 = 26, \ldots$$

But it may not be easy to write down the general term.

**Recall:** A sequence of real numbers is a function $a : \mathbb{Z}^+ \to \mathbb{R}$.

### Question

Why does a recursive relation defines a sequence?

The next two slides are optional.

**Theorem 5.1 (Recursion Theorem)**

*Let $X$ be a set, $b \in X$, and $f : X \to X$ be a function. Then there is a unique function $a : \mathbb{Z}^+ \to X$ such that*

$$a(1) = b, \quad \text{and}$$
$$a(n+1) = f(a(n)) \quad \text{for all } n \in \mathbb{Z}^+.$$

**Example 5.2:** $a_1 = 4$ and $a_{n+1} = a_n + 1$ for $n \geq 1$.
Apply Recursion Theorem to

$$b = 4 \quad \text{and} \quad f(x) = x + 1.$$

**Example 5.4:** $a_1 = 1$ and $a_{n+1} = a_n^2 + 1$ for $n \geq 1$.
Apply Recursion Theorem to

$$b = 1 \quad \text{and } f(x) = x^2 + 1.$$

**Theorem 5.2 (Recursion Theorem)**

*Let $X$ be a set, $b \in X$, and $f : X \to X$ be a function. Then there is a unique function $a : \mathbb{Z}^+ \to X$ such that*

$$a(1) = b, \quad \text{and}$$
$$a(n+1) = f(a(n)) \quad \text{for all } n \in \mathbb{Z}^+.$$

**Example 5.3:** $F_1 = F_2 = 1$ and $F_{n+2} = F_{n+1} + F_n$ for $n \geq 1$.
Apply Recursion Theorem to

$$b = (1,1) \quad \text{and } f(x,y) = (y, x+y).$$

Then we obtain a sequence $a(n) := (F_n, G_n)$ such that $a(1) = (1,1)$ and

$$\begin{pmatrix} F_{n+1} \\ G_{n+1} \end{pmatrix} = a(n+1) = f(a(n)) = \begin{pmatrix} G_n \\ F_n + G_n \end{pmatrix},$$

So $G_n = F_{n+1}$ and hence

$$F_{n+2} = G_{n+1} = F_n + G_n = F_n + F_{n+1}.$$

**Definition 5.2 (Arithmetic sequence)**

A sequence $\{a_n\}$ is said to be an **arithmetic sequence** (or arithmetic progression) if there is a constant $d$ such that

$$a_{n+1} - a_n = d \quad \text{for any } n \in \mathbb{Z}^+.$$

**Remark:**

(1) The number $d$ is unique ($\because d = a_2 - a_1$), and is called the **common difference** of $\{a_n\}$.

(2) The general term is

$$a_n = a + (n-1)d \quad \text{for any } n \in \mathbb{Z}^+,$$

where $a = a_1$ is the **initial term**.

Consider the arithmetic series $s_n = \sum_{k=1}^{n} a_k = \sum_{k=1}^{n} [a + (k-1)d]$.

Goal: simplify the sum.

$$\text{Let } j = n + 1 - k. \text{ Then } k - 1 = n - j.$$
$$\text{When } k = 1, j = n; \text{ when } k = n, j = 1.$$

So,

$$s_n = \sum_{j=1}^{n} [a + (n-j)d] = \sum_{k=1}^{n} [a + (n-k)d] \,.$$

Then,

$$\begin{aligned} 2s_n = s_n + s_n &= \sum_{k=1}^{n} [a + (k-1)d] + \sum_{k=1}^{n} [a + (n-k)d] \\ &= \sum_{k=1}^{n} [2a + (n-1)d] = n\,[2a + (n-1)d] \end{aligned}$$

Hence

$$s_n = \frac{n}{2}\,[2a + (n-1)d] \;= \frac{n}{2}(a_1 + a_n).$$

## Example 5.5

The common difference and the sum of the first 15 terms of an arithmetic sequence are $\frac{3}{2}$ and 240 respectively.

(a) Find the first term of the sequence.

(b) If the sum of the first $k$ terms of the sequence is 361, find the value of $k$.

(a) Recall:

$$s_n = \frac{n}{2}\left[2a + (n-1)d\right].$$

Since $d = \frac{3}{2}$ and $s_{15} = \dfrac{15}{2}[2a + (15-1)\dfrac{3}{2}] = 240$, we have $a = 5.5$.

(b) Given:

$$s_k = \frac{k}{2}\left[2(5.5) + (k-1)(\frac{3}{2})\right] = 361,$$

we have

$$3k^2 + 19k - 1444 = 0.$$

So $k = 19$ ($k = -76/3$ is rejected).

**Definition 5.3 (Geometric sequence)**

A sequence $\{b_n\}$ of nonzero numbers is said to be a **geometric sequence** (or geometric progression) if there is a nonzero constant $r$ such that

$$\frac{b_{n+1}}{b_n} = r \quad \text{for any } n \in \mathbb{Z}^+.$$

**Remark:**

(1) The number $r$ is unique ($\because r = b_2/b_1$), and is called the **common ratio** of $\{b_n\}$.

(2) The general term is

$$b_n = br^{n-1} \quad \text{for any } n \in \mathbb{Z}^+,$$

where $b = b_1$ is the **initial term**.

# Arithmetic and Geometric Sequences

Geometric series $s_n = \sum_{k=1}^{n} b_k = \sum_{k=1}^{n} br^{k-1} = b\sum_{k=0}^{n-1} r^k$. (How to simplify?)

If $r = 1$, $s_n = bn$ (simple).

If $r \neq 1$, then

$$\begin{aligned}(1-r)\left(\sum_{k=0}^{n-1} r^k\right) &= \sum_{k=0}^{n-1} r^k - \sum_{k=0}^{n-1} r^{k+1} \\ &= \sum_{k=0}^{n-1} r^k - \sum_{k=1}^{n} r^k \\ &= 1 - r^n.\end{aligned}$$

So

$$\sum_{k=0}^{n-1} r^k = \frac{1-r^n}{1-r},$$

and

$$s_n = \frac{b(1-r^n)}{1-r} = \frac{b(r^n-1)}{r-1}.$$

## Example 5.6 (Mortgage Formula)

$P$ = Mortgage principal
$R$ = Fixed interest rate
$N$ = Mortgage duration (in month)
$x$ = Monthly payment amount

| Month | 0 | 1 | 2 | $\cdots$ |
|---|---|---|---|---|
| Loan Amount | $P$ | $P(1+\frac{R}{12})-x$ | $[P(1+\frac{R}{12})-x](1+\frac{R}{12})-x$ | $\cdots$ |

Given $P$, $R$ and $N$, find $x$.

Let $L_n$ be the loan amount after $n$-th month (after monthly payment). Then

$$L_0 = P \quad \text{and} \quad L_n = L_{n-1}(1+\frac{R}{12}) - x \quad \text{for } n \geq 1.$$

Hence,

$$\begin{aligned} L_n &= P(1+\frac{R}{12})^n - \left[(1+\frac{R}{12})^{n-1}x + (1+\frac{R}{12})^{n-2}x + \cdots + (1+\frac{R}{12})x + x\right] \\ &= P(1+\frac{R}{12})^n - \frac{(1+\frac{R}{12})^n - 1}{(1+\frac{R}{12}) - 1}x. \end{aligned}$$

Hence,

$$L_n = P(1+\frac{R}{12})^n - \frac{(1+\frac{R}{12})^n - 1}{(1+\frac{R}{12}) - 1}x.$$

Set $L_N = 0$, we have

$$P(1+\frac{R}{12})^N = \frac{(1+\frac{R}{12})^N - 1}{(1+\frac{R}{12}) - 1}x$$

$$x = P\frac{\frac{R}{12}(1+\frac{R}{12})^N}{(1+\frac{R}{12})^N - 1}.$$

For example,

$$\text{if } P = 8,000,000,\ R = 3\%,\ N = 360, \text{ then } x = 33,728.$$

(Total interest $= 4,142,196$.)
GeoGebra Illustration: Mortage

## Exercise 5.3

Let $(x_n)$ be an arithmetic-geometric sequence defined by

$$x_n = (a + nd)br^n, \quad n \in \mathbb{N}, \qquad \text{where } a, d, r \in \mathbb{R} \text{ and } r \neq 0, \neq 1.$$

Show that $S_n := \sum_{k=0}^{n-1} x_k$ can be expressed as

$$S_n = \frac{ab - (a + nd)br^n}{1 - r} + \frac{dbr(1 - r^n)}{(1 - r)^2}.$$

**Hint:** Consider $S_n - rS_n$ .